

index, i, j, n, m indices

terminals	$ \begin{array}{l} ::= \\ \quad \longrightarrow \\ \quad \rightarrow \\ \quad \Rightarrow \\ \quad \lambda \\ \quad \mapsto \\ \quad \vdash \\ \quad \emptyset \\ \quad \times \\ \quad <: \\ \quad \langle \\ \quad \rangle \end{array} $	
Γ	$ \begin{array}{l} ::= \\ \quad \emptyset \\ \quad \Gamma, x : T \end{array} $	<i>contexts:</i> <i>empty context</i> <i>term variable binding</i>
formula	$ \begin{array}{l} ::= \\ \quad \text{judgement} \\ \quad x : T \in \Gamma \\ \quad \text{formula}_1 \dots \text{formula}_n \\ \quad j \text{ INDEXES } t_1 \dots t_n \\ \quad j \text{ INDEXES } T_1 \dots T_n \end{array} $	
T	$ \begin{array}{l} ::= \\ \quad \text{Unit} \\ \quad \{ T_i^{i \in 1..n} \} \end{array} $	<i>types:</i> <i>unit type</i> <i>tuple type</i>
t	$ \begin{array}{l} ::= \\ \quad (t) \\ \quad \text{unit} \\ \quad \{ t_i^{i \in 1..n} \} \\ \quad t.i \end{array} $	<i>terms:</i> M <i>constant unit</i> <i>tuple</i> <i>projection</i>
v	$ \begin{array}{l} ::= \\ \quad \text{unit} \\ \quad \{ v_i^{i \in 1..n} \} \end{array} $	<i>values:</i> <i>unit value</i> <i>tuple value</i>
Jop	$ \begin{array}{l} ::= \\ \quad t \longrightarrow t' \end{array} $	<i>Evaluation</i>
Jtype	$ \begin{array}{l} ::= \\ \quad \Gamma \vdash t : T \end{array} $	<i>Typing</i>
judgement	$ \begin{array}{l} ::= \\ \quad \text{Jop} \\ \quad \text{Jtype} \end{array} $	
user_syntax	$::= $	

	<i>index</i>
	terminals
	Γ
	formula
	T
	t
	v

$\boxed{t \longrightarrow t'}$ *Evaluation*

$$\frac{j \text{ INDEXES } v_i^{i \in 1..n}}{\{v_i^{i \in 1..n}\}.j \longrightarrow v_j} \quad \text{E_PROJTUPLE}$$

$$\frac{t_1 \longrightarrow t'_1}{t_1.i \longrightarrow t'_1.i} \quad \text{E_PROJTP}$$

$$\frac{t \longrightarrow t'}{\{v_i^{i \in 1..m}, t, t_j^{j \in 1..n}\} \longrightarrow \{v_i^{i \in 1..m}, t', t_j^{j \in 1..n}\}} \quad \text{E_TUPLE}$$

$\boxed{\Gamma \vdash t : T}$ *Typing*

$$\frac{}{\Gamma \vdash \text{unit} : \text{Unit}} \quad \text{T_UNIT}$$

$$\frac{\Gamma \vdash t_i : T_i^{i \in 1..n}}{\Gamma \vdash \{t_i^{i \in 1..n}\} : \{T_i^{i \in 1..n}\}} \quad \text{T_TUPLE}$$

$$\frac{\Gamma \vdash t_1 : \{T_i^{i \in 1..n}\} \quad j \text{ INDEXES } T_i^{i \in 1..n}}{\Gamma \vdash t_1.j : T_j} \quad \text{T_PROJTP}$$

Definition rules: 6 good 0 bad
Definition rule clauses: 12 good 0 bad